

Esempio ($n=1$)

$$1) \quad T = \delta_0 \Rightarrow \boxed{\widehat{\delta_0} = 1}$$

$$\begin{aligned} \langle \widehat{\delta_0}, \varphi \rangle_{(\mathcal{S}'(\mathbb{R}^n), \mathcal{S}(\mathbb{R}^n))} &= \langle \delta_0, \widehat{\varphi} \rangle_{(\mathcal{S}'(\mathbb{R}^n), \mathcal{S}(\mathbb{R}^n))} = \widehat{\varphi}(0) = \\ &= \int_{\mathbb{R}} \varphi(x) dx = \langle 1, \varphi \rangle_{(\mathcal{S}'(\mathbb{R}^n), \mathcal{S}(\mathbb{R}^n))} \end{aligned}$$

$$\left[\widehat{\varphi}(\xi) = \int_{\mathbb{R}} \varphi(x) e^{-i\xi x} dx \Rightarrow \widehat{\varphi}(0) = \int_{\mathbb{R}} \varphi(x) dx \right]$$

$$2) \quad \widehat{1} = \widehat{\delta_0} = (2\pi) \check{\delta_0} = (2\pi) \delta_0$$

$$\langle \check{\delta_0}, \varphi \rangle = \langle \delta_0, \check{\varphi} \rangle \stackrel{\text{per def}}{=} \langle \delta_0, \varphi \rangle$$

$$\boxed{\widehat{1} = (2\pi) \delta_0}$$

$$\check{\varphi}(x) \stackrel{\text{per def}}{=} \varphi(-x)$$

$$3) \quad \widehat{x} = \mathcal{F}(x \cdot 1) \stackrel{\uparrow}{=} i(\widehat{1})' = (2\pi i)(\delta_0)'$$

derivata di trasformatà

$$\langle (\delta_0)', \varphi \rangle = - \langle \delta_0, \varphi' \rangle$$

$$\boxed{\widehat{x} = (2\pi i)(\delta_0)'}$$

Esempi di applicazione di $\mathcal{F}$ alla risoluzione di PDE

Eq. lineari del 2° ordine $\begin{cases} \nearrow \text{calore} & (\text{parabolico}) \\ \searrow \text{onde} & (\text{iperbolico}). \end{cases}$

Eq. calore

$$u = u(x_1, \dots, x_n, t)$$

$$\begin{cases} u_t - \Delta u = f & x \in \mathbb{R}^n, \quad t \in (0, +\infty) \\ u(x, 0) = u_0(x) \end{cases}$$

$\Delta u = u_{x_1 x_1} + \dots + u_{x_n x_n}$

Eq. onde

$$\begin{cases} u_{tt} - \Delta u = f & x \in \mathbb{R}^n, \quad t \in (0, +\infty) \\ u(x, 0) = u_0(x) & x \in \mathbb{R}^n \\ u_t(x, 0) = u_1(x) & x \in \mathbb{R}^n \end{cases}$$

Calore

$$\begin{cases} u_t - u_{xx} = 0 & x \in \mathbb{R}, \quad t \in (0, +\infty) \\ u(x, 0) = u_0(x) \in L^2(\mathbb{R}) \end{cases}$$

Strategia: applicare $\mathcal{F}_{x \rightarrow \xi}(u)$

$$\hat{u}(\xi, t) = \int_{\mathbb{R}} u(x, t) e^{-i\xi x} dx$$

$$\hat{u}(\xi, 0) = \int_{\mathbb{R}} u(x, 0) e^{-i\xi x} dx = \int_{\mathbb{R}} u_0(x) e^{-i\xi x} dx = \hat{u}_0(\xi)$$

$$(1) \bullet \widehat{u_t} = (\hat{u})_t$$

$$\left(\frac{\partial}{\partial t} \right) \left(\int_{\mathbb{R}} u(x, t) e^{-i\xi x} dx \right) = \int_{\mathbb{R}} u_t(x, t) e^{-i\xi x} dx$$

$$\bullet \widehat{u_{xx}} = (i\xi)^2 \hat{u} = -\xi^2 \hat{u}$$

$$\begin{cases} (\hat{u})_t + \xi^2 \hat{u} = 0 \\ \hat{u}(\xi, 0) = \hat{u}_0(\xi) \end{cases}$$

$$\hat{u} = \hat{u}(\xi, t)$$

$$(2) \quad \varphi(t) = \hat{u}(\xi, t)$$

$$\begin{cases} \varphi'(t) = -\xi^2 \varphi(t) \\ \varphi(0) = \hat{u}_0(\xi) \end{cases} \Rightarrow \varphi(t) = C \cdot e^{-\xi^2 t} = \hat{u}_0(\xi) e^{-\xi^2 t}$$

$$C = \varphi(0) = \hat{u}_0(\xi)$$

$$\hat{u}(\xi, t) = \hat{u}_0(\xi) \cdot e^{-\xi^2 t} = \mathcal{F}(u_0) \cdot \mathcal{F}(\text{?})$$

$$\textcircled{3} \quad u(x, t) = u_0 \otimes \mathcal{F}^{-1}(e^{-\xi^2 t})$$

$$\mathcal{F}^{-1}(e^{-\xi^2 t}) = \frac{1}{\sqrt{4\pi t}} e^{-\frac{x^2}{4t}}$$

$$\mathcal{F}\left(\frac{1}{\sqrt{4\pi t}} e^{-\frac{x^2}{4t}}\right) = e^{-\xi^2 t}$$

segue dalla trasformata di e^{-x^2}
e dalla regola di trasformazioni per riscalamento
scrivendo $\frac{x^2}{4t} = \left(\frac{x}{\sqrt{4t}}\right)^2$

Conclusione:

$$u(x, t) = u_0 \otimes_x \left(\frac{1}{\sqrt{4\pi t}} e^{-\frac{x^2}{4t}} \right)$$

$$u(x, t) = \frac{1}{\sqrt{4\pi t}} \int_{\mathbb{R}} u_0(x-y) e^{-\frac{y^2}{4t}} dy$$

Commenti

- 1) prodotto di convoluzione ha senso perché $u_0 \in L^2(\mathbb{R})$ e la gaussiana $\in L^1(\mathbb{R})$. (e sappiamo $L^2(\mathbb{R}) * L^1(\mathbb{R}) \subseteq L^2(\mathbb{R})$)
- $\lim_{t \rightarrow 0} u(x, t) = u_0(x)$ in $L^2(\mathbb{R})$

$$\|u(x, t) - u_0(x)\|_{L^2}^2 = \frac{1}{2\pi} \|\hat{u}(\xi, t) - \hat{u}_0(\xi)\|_{L^2}^2$$

$$= \frac{1}{2\pi} \|\hat{u}_0(\xi) e^{-\xi^2 t} - \hat{u}_0(\xi)\|_{L^2}^2$$

$$= \frac{1}{2\pi} \int_{\mathbb{R}} \underbrace{|\hat{u}_0(\xi)|^2 |e^{-\xi^2 t_m} - 1|^2}_{f_m(\xi)} d\xi \xrightarrow[t_m \rightarrow 0^+]{(?) } 0$$

$$f_m(\xi) \rightarrow 0 \quad (\text{perché } (e^{-\xi^2 t_m} - 1) \rightarrow 0)$$

$$|f_m(\xi)| \leq 4 |\hat{u}_0(\xi)|^2 \in L^1(\mathbb{R}) \quad \left(\begin{array}{l} \text{perché } u_0 \in L^2 \\ \Rightarrow \hat{u}_0 \in L^2 \end{array} \right)$$

- Effetto regolarizzante dell'eq. del calore $\partial_t - \Delta_x$

$$\forall \bar{t} > 0, \quad x \mapsto u(x, \bar{t}) \text{ è di classe } C^\infty(\mathbb{R})$$

- Velocità di propagazione dell'eq. calore è infinita.

$$\forall \bar{t} > 0, \quad x \mapsto u(x, \bar{t}) \text{ è diversa da zero su tutto } \mathbb{R}$$

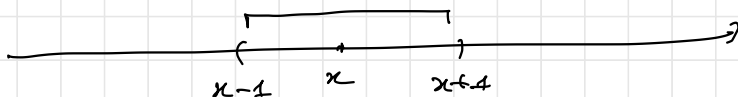
anche se $u_0(x)$ è a supporto compatto

Esempio

$$u_0(x) = \chi_{(-1,1)}(x)$$

$$u(x,t) = \int_{\mathbb{R}} \chi_{(-1,1)}(x-y) e^{-\frac{y^2}{4t}} dy$$

$$\begin{aligned} &= \int_{x-1}^{x+1} e^{-\frac{y^2}{4t}} dy > 0 \\ &\quad \begin{array}{l} \uparrow \\ -1 \leq x-y \leq 1 \\ x-1 \leq y \leq x+1 \end{array} \end{aligned}$$



Onale

$$(P) \begin{cases} u_{tt} - u_{xx} = 0 & x \in \mathbb{R}, t \in (0, +\infty) \\ u(x, 0) = u_0(x) \\ u_t(x, 0) = u_1(x) \end{cases}$$

$$\text{Ip. } u_0 \in \mathcal{C}^2(\mathbb{R}) \cap L^1(\mathbb{R}), \quad u_1 \in \mathcal{C}^1(\mathbb{R}) \cap L^1(\mathbb{R})$$

Strategia: applicare $\mathcal{F}_{x \rightarrow \xi}(u)$

$$\hat{u}(\xi, t) = \int_{\mathbb{R}} u(x, t) e^{-i\xi x} dx$$

$$\hat{u}(\xi, 0) = \int_{\mathbb{R}} u(x, 0) e^{-i\xi x} dx = \int_{\mathbb{R}} u_0(x) e^{-i\xi x} dx = \hat{u}_0(\xi)$$

$$\hat{u}_t(\xi, 0) = \int_{\mathbb{R}} u_t(x, 0) e^{-i\xi x} dx = \int_{\mathbb{R}} u_1(x) e^{-i\xi x} dx = \hat{u}_1(\xi)$$

$$(P_1) \begin{cases} v_{tt} - v_{xx} = 0 \\ v(x, 0) = u_0(x) \\ v_t(x, 0) = 0 \end{cases} \quad (P_2) \begin{cases} w_{tt} - w_{xx} = 0 \\ w(x, 0) = 0 \\ w_t(x, 0) = u_1(x). \end{cases}$$

Dim. Se v resolve (P_1) e w resolve (P_2)

$$\Rightarrow u = v + w \text{ resolve } (P)$$

Solution 1 (P1)

$$\textcircled{1} \quad \begin{cases} (\hat{v})_{tt} + \xi^2 \hat{v} = 0 \\ \hat{v}(\xi, 0) = \hat{u}_0(\xi) \\ \hat{v}_t(\xi, 0) = 0 \end{cases}$$

$$\textcircled{2} \quad \varphi(t) := \hat{v}(\xi, t)$$

$$\begin{cases} \varphi''(t) + \xi^2 \varphi(t) = 0 & \lambda^2 + \xi^2 = 0 \Rightarrow \lambda = \pm i\xi \\ \varphi(0) = \hat{u}_0(\xi) \\ \varphi'(0) = 0 \end{cases}$$

$$\varphi(t) = C_1 e^{i\xi t} + C_2 e^{-i\xi t}$$

$$\varphi(0) = C_1 + C_2 = \hat{u}_0(\xi)$$

$$\varphi'(t) = C_1 i\xi e^{i\xi t} - C_2 i\xi e^{-i\xi t}$$

$$\varphi'(0) = i\xi (C_1 - C_2) = 0 \Rightarrow C_1 = C_2$$

$$\Rightarrow C_1 = C_2 = \frac{1}{2} \hat{u}_0(\xi)$$

$$\Rightarrow \varphi(t) = \frac{1}{2} \hat{u}_0(\xi) e^{i\xi t} + \frac{1}{2} \hat{u}_0(\xi) e^{-i\xi t}$$

$$\Rightarrow \hat{v}(\xi, t) = \frac{1}{2} \hat{u}_0(\xi) e^{i\xi t} + \frac{1}{2} \hat{u}_0(\xi) e^{-i\xi t}$$

$$\textcircled{3} \quad v(x, t) = \frac{1}{2} [\mu_0(x+t) + \mu_0(x-t)]$$

Solution d. (P₂)

$$\textcircled{1} \quad \begin{cases} (\hat{w})_{tt} + \xi^2 \hat{w} = 0 \\ \hat{w}(\xi, 0) = 0 \\ (\hat{w})_t(\xi, 0) = \hat{u}_1(\xi) \end{cases}$$

$$\textcircled{2} \quad \psi(t) := \hat{w}(\xi, t)$$
$$\begin{cases} \psi'' + \xi^2 \psi = 0 \\ \psi(0) = 0 \\ \psi'(0) = \hat{u}_1(\xi) \end{cases}$$

$$\psi(t) = D_1 \cos(\xi t) + D_2 \sin(\xi t)$$

$$\psi(0) = 0 \Rightarrow D_1 = 0$$

$$\psi(t) = D_2 \sin(\xi t)$$

$$\psi'(t) = D_2 \xi \cos(\xi t)$$

$$\psi'(0) = \hat{u}_1(\xi) \iff D_2 \xi = \hat{u}_1(\xi) \iff D_2 = \frac{\hat{u}_1(\xi)}{\xi}$$

$$\psi(t) = \hat{w}(\xi, t) = \frac{\hat{u}_1(\xi)}{\xi} \sin(\xi t) = \hat{u}_1(\xi) \cdot \left(\frac{\sin(\xi t)}{\xi} \right)$$

$$\textcircled{3} \quad w(x, t) = u_1 *_{\xi} \underbrace{\int^{-1} \left(\frac{\sin(\xi t)}{\xi} \right)}_{\textcircled{?}}$$

$$p(x) = \chi_{(-1,1)}(x) \xrightarrow{\mathcal{F}} 2 \frac{\sin \xi}{\xi}$$

$$\frac{1}{2} p\left(\frac{x}{t}\right) \xrightarrow{\mathcal{F}} \frac{\sin(\xi t)}{\xi}$$

$$w(x,t) = u_1 * \frac{1}{2} p\left(\frac{x}{t}\right)$$

$$= \frac{1}{2} \int_{\mathbb{R}} p\left(\frac{x-y}{t}\right) u_1(y) dy$$

$$\uparrow$$

$$= \frac{1}{2} \int_{x-t}^{x+t} u_1(y) dy$$

$$-1 \leq \frac{x-y}{t} \leq 1$$

$$x-t \leq y \leq x+t$$

$$w(x,t) = \frac{1}{2} \int_{x-t}^{x+t} u_1(y) dy \underset{U_1 \text{ prim. di } u_1}{=} \frac{1}{2} [U_1(x+t) - U_1(x-t)]$$

La formula per la sol. (P) è (detta di D'Alembert)

$$u(x,t) = \frac{1}{2} \left[u_0(x+t) + u_0(x-t) \right] + \frac{1}{2} \int_{x-t}^{x+t} u_1(y) dy$$

Commenti:

- Se $u_0 \in \mathcal{C}^2(\mathbb{R})$, $u_1 \in \mathcal{C}^1(\mathbb{R}) \Rightarrow u(x,t) \in \mathcal{C}^2$ che soddisfa l'eq. (in senso classico).

non c'è effetto regolarizzante.

- La velocità di propagazione è finita

Esempio $u_0(x) = \chi_{(-1,1)}(x)$ $u_1(x) \equiv 0$.

$\forall \bar{t} > 0$, $x \mapsto u(x, \bar{t})$ è a supporto compatto

